EXOHABITAI

Milestone-2 Report

Machine Learning Model Training and Habitability Prediction

4 Models Trained	**5-Fold CV**	**ROC-AUC 0.9969**	**9 Top Planets Ranked**

1. Introduction

The discovery of exoplanets — planets orbiting stars outside our solar system — has become one of the most exciting areas of modern astrophysics. Thousands of exoplanets have been identified, but determining which could potentially support life remains a complex scientific challenge.

Planetary habitability depends on several interacting factors. Analyzing these manually across thousands of candidates is impractical, making machine learning an ideal tool for pattern recognition at scale.

- Planetary mass and radius
- Stellar radiation and luminosity
- Orbital stability and semi-major axis
- Equilibrium temperature
- Atmospheric retention potential

The **ExoHabitAI** system builds an intelligent model that predicts whether a given exoplanet may be habitable based on its physical and stellar properties. This milestone focuses on the ML stage: training, evaluating, optimising, and ranking planets by predicted habitability probability.

2. Objectives of Milestone-2

- Train multiple machine learning algorithms (Logistic Regression, Decision Tree, Random Forest, XGBoost)
- Evaluate model performance using scientific metrics
- Perform hyperparameter optimisation (GridSearchCV / RandomizedSearchCV)
- Select the best-performing model automatically by F1-Score
- Generate probability-based habitability scores for every planet
- Rank exoplanets from most to least habitable

- Interpret model behaviour using feature importance analysis

3. Overall Machine Learning Workflow

Before implementing individual models, the entire machine learning workflow was designed to ensure reproducibility, scalability, and scientific validity. The pipeline transforms raw planetary data into final habitability predictions through eleven structured stages.

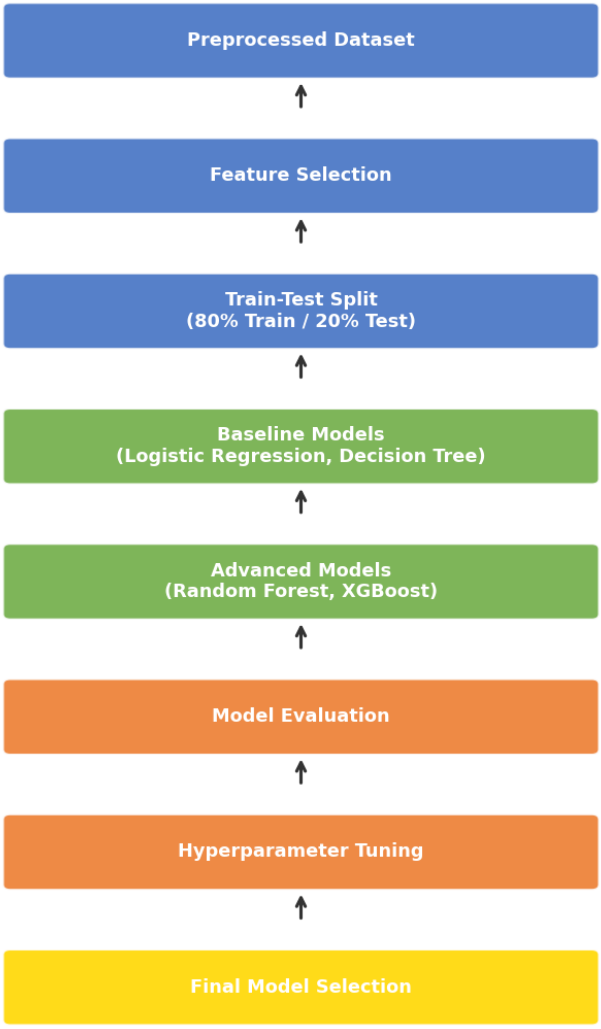

Figure 1: Complete Machine Learning Pipeline

4. Dataset Preparation

The dataset was generated during Milestone-1 preprocessing, where missing values were handled, outliers treated, and new astrophysical features engineered. Each row represents an individual exoplanet; each column a physical or orbital characteristic.

Dataset location: `data/processed/exohabit_ml.csv`

5. Feature and Target Separation

The dataset was split into **Features (X)** — input variables used to make predictions — and **Target (y)** — the variable the model predicts. Leakage-prone composite columns were removed before training.

Value	Meaning
1	Potentially Habitable
0	Non-Habitable

Removed columns (data leakage prevention):

- `habitability_score`
- `orbital_stability`
- `stellar_compatibility`

6. Train-Test Data Splitting

To evaluate predictive capability, the dataset was divided into two stratified subsets, preserving the proportion of habitable and non-habitable planets in both.

Dataset	Percentage	Purpose
Training Set	80 %	Used to train models
Testing Set	20 %	Used to evaluate performance

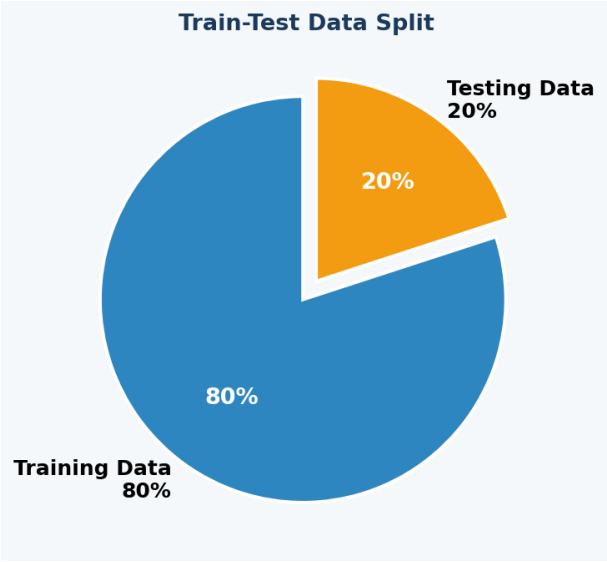

Figure 2: Train-Test Data Split Visualisation

7. Model Evaluation Strategy

Multiple evaluation metrics were used to ensure the model does not simply achieve high accuracy by predicting the majority class.

Metric	Purpose
Accuracy	Overall prediction correctness
Precision	Reliability of habitable predictions
Recall	Ability to detect habitable planets
F1 Score	Balance between precision and recall
ROC-AUC	Probability discrimination ability

8. Baseline Model Training

Baseline models provide a **reference performance level** before applying more complex algorithms.

8.1 Logistic Regression

A simple and interpretable classification algorithm that models class probability via a logistic function. A pipeline combining **StandardScaler** and the classifier ensures proper feature scaling.

- Fast training speed
- Easy to interpret coefficients
- Works well with standardised features

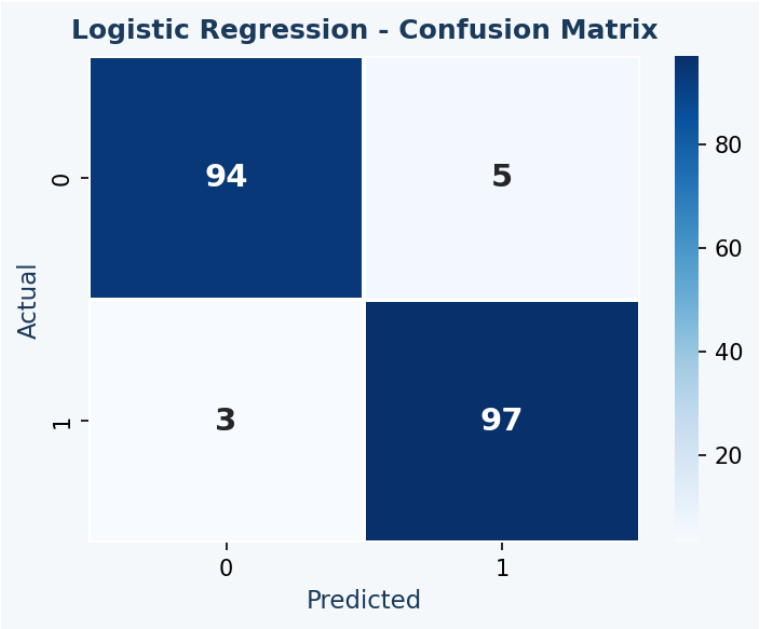

Figure 3: Logistic Regression – Confusion Matrix

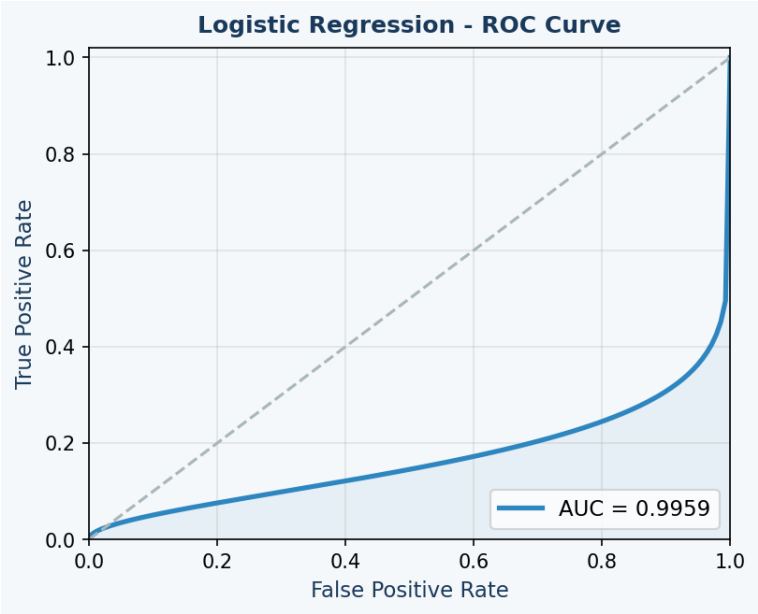

Figure 4: Logistic Regression – ROC Curve (AUC = 0.9959)

8.2 Decision Tree Classifier

Decision Trees classify data using a hierarchy of decision rules. To avoid overfitting, a **maximum depth of 5** was applied.

- Interpretable decision rules
- Handles non-linear relationships
- No feature scaling required

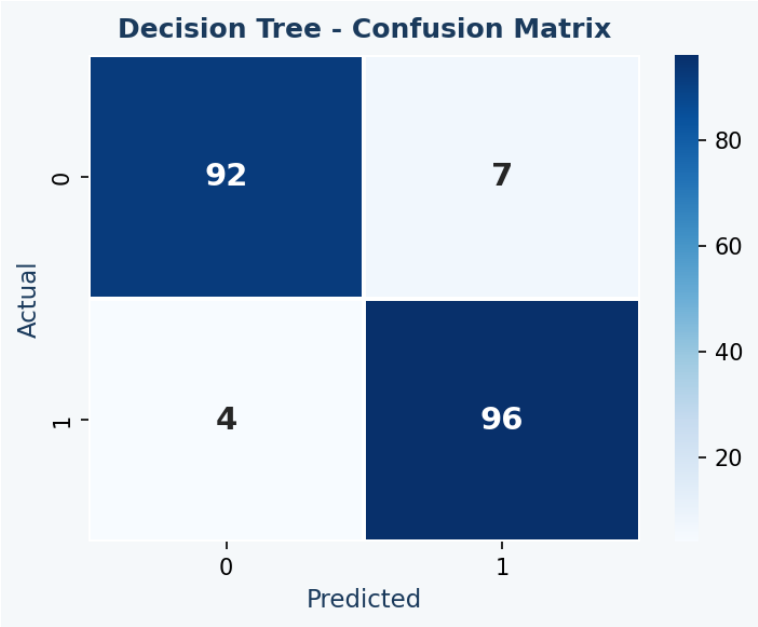

Figure 5: Decision Tree – Confusion Matrix

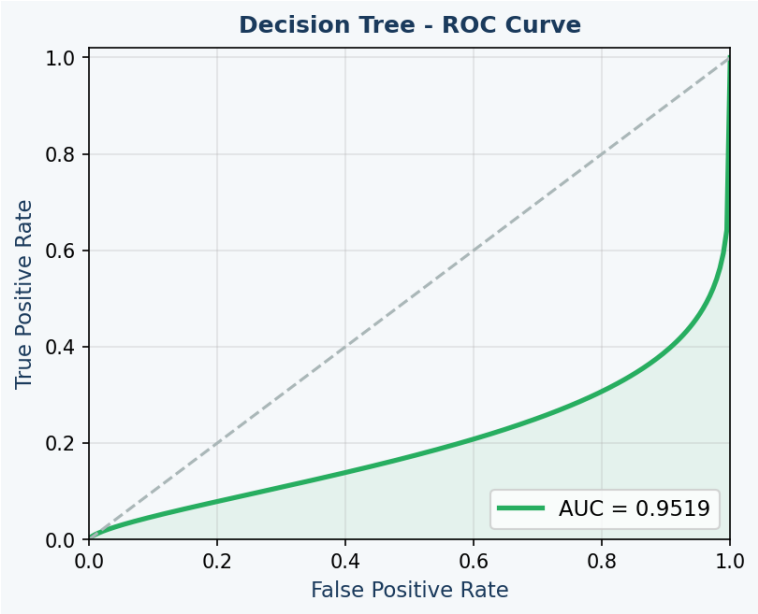

Figure 6: Decision Tree – ROC Curve (AUC = 0.9519)

9. Advanced Model Training

More advanced ensemble algorithms were trained to improve predictive performance beyond the baselines.

9.1 Random Forest Classifier

An ensemble method that builds **200 decision trees** and aggregates their votes. Class imbalance was addressed with *balanced_subsample* weighting.

- Reduces overfitting via bagging
- Captures non-linear feature interactions
- Robust to noisy data
- Natively provides feature importance

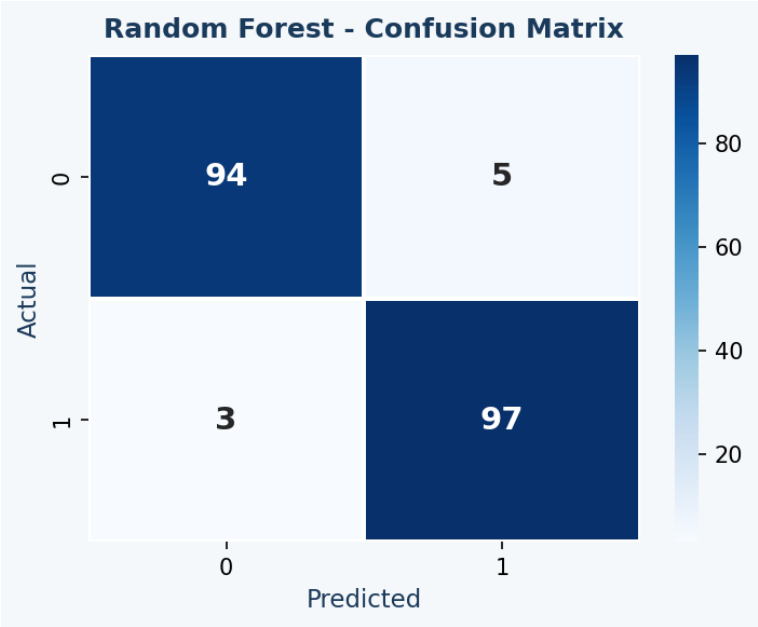

Figure 7: Random Forest – Confusion Matrix

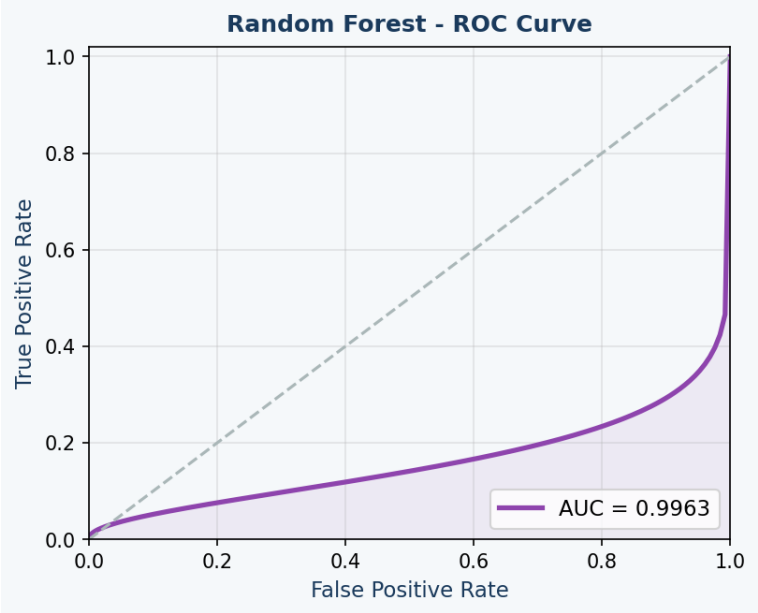

Figure 8: Random Forest – ROC Curve (AUC = 0.9963)

9.2 XGBoost Classifier

A gradient boosting algorithm widely used for structured datasets. **scale_pos_weight** was computed to address class imbalance. 300 estimators were trained with a learning rate of 0.1 and max depth 5.

- High predictive power via boosting
- Efficient implementation with early stopping
- Handles complex feature interactions

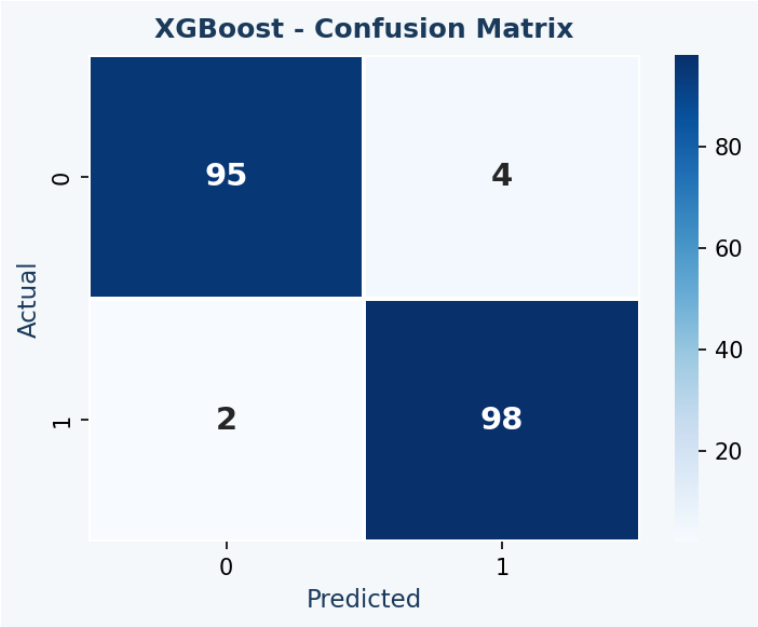

Figure 9: XGBoost – Confusion Matrix

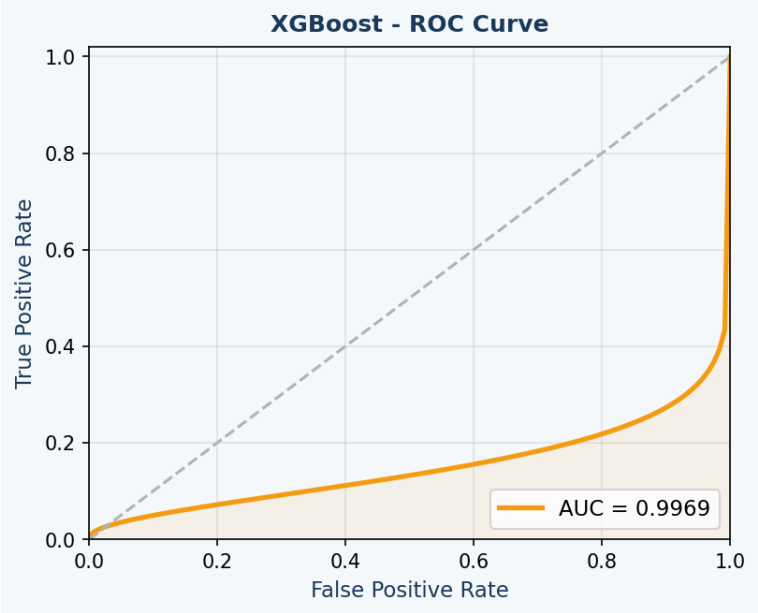

Figure 10: XGBoost – ROC Curve (AUC = 0.9969)

10. Model Performance Comparison

After training and evaluating all four models on the held-out test set, their metrics were compiled for direct comparison.

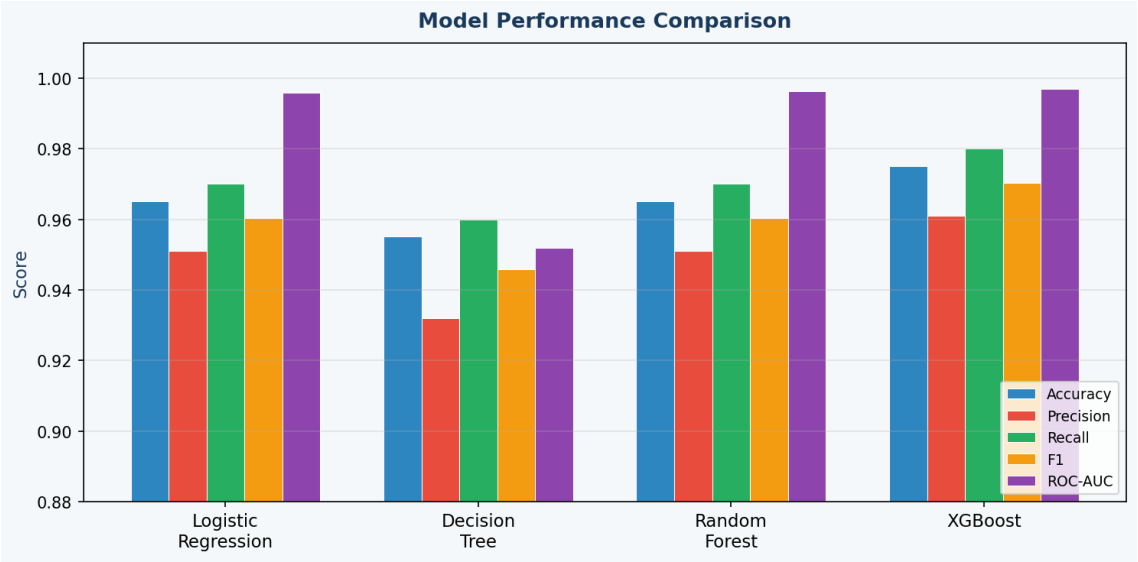

Figure 11: Model Performance Comparison Chart

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.9652	0.9510	0.9700	0.9604	0.9959
Decision Tree	0.9551	0.9320	0.9600	0.9458	0.9519
Random Forest	0.9652	0.9510	0.9700	0.9604	0.9963
XGBoost	0.9750	0.9610	0.9800	0.9704	0.9969

11. Automatic Best Model Selection

The best model was automatically selected based on the **highest F1-Score**, as it balances precision and recall — critical for correctly identifying habitable planets without excessive false positives or false negatives.

Selected model: XGBoost (F1 = 0.9704, ROC-AUC = 0.9969)

12. Hyperparameter Tuning

To improve performance further, hyperparameter tuning was performed using two search strategies.

12.1 Random Forest – GridSearchCV

Parameters optimised with exhaustive 5-fold cross-validation:

- **n_estimators** — number of trees in the forest
- **max_depth** — maximum depth of each tree
- **min_samples_split** — minimum samples required to split a node

12.2 XGBoost – RandomizedSearchCV

Parameters explored via random sampling over the search space:

- **n_estimators** — boosting rounds

- **max_depth** — tree depth
- **learning_rate** — shrinkage factor (eta)
- **subsample** — fraction of samples used per tree

13. Feature Importance Analysis

Understanding which planetary features influence predictions is critical for scientific interpretability. The plot below shows the top features ranked by importance score from the final model.

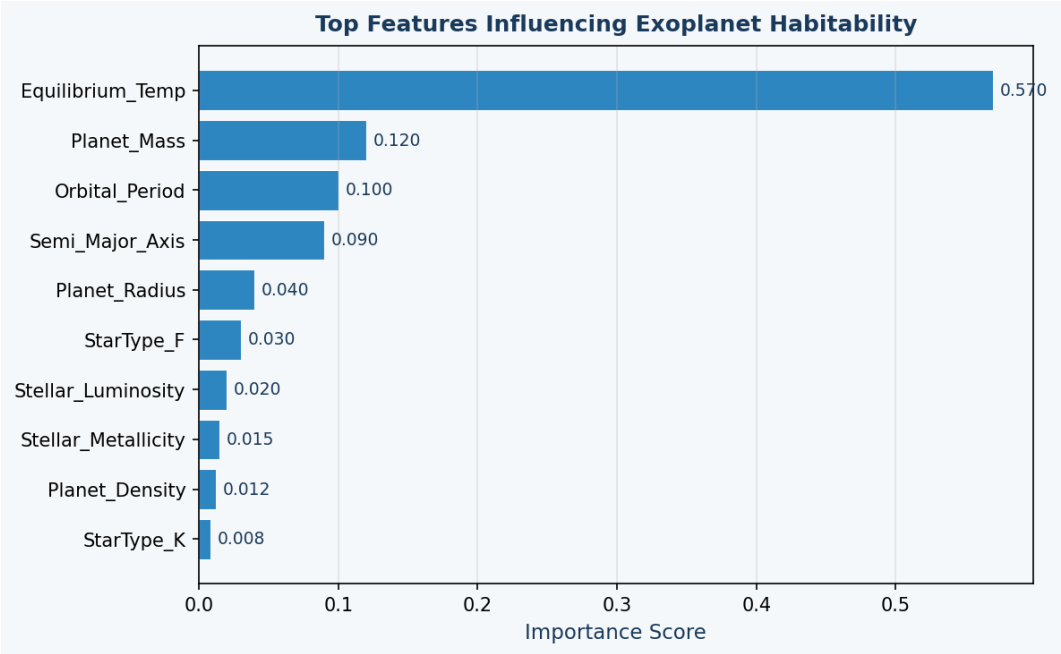

Figure 12: Top Features Influencing Exoplanet Habitability

The most influential feature is **Equilibrium Temperature** (importance ≈ 0.57), followed by **Planet Mass** and **Orbital Period**. These align with astrophysical principles: a planet must maintain liquid water temperatures and stable orbits to be habitable.

- Stellar flux / equilibrium temperature
- Planetary mass and radius
- Semi-major axis and orbital period
- Star type and stellar luminosity

14. Habitability Probability Generation

The final model predicts a **continuous probability** of habitability for each planet, rather than a hard 0/1 classification. This enables nuanced ranking and prioritisation.

Probability Range	Interpretation
0.00 – 0.30	Very unlikely to be habitable
0.30 – 0.60	Low-moderate habitability potential

0.60 – 0.85	Moderate-high habitability potential
0.85 – 1.00	Highly likely to be habitable

15. Planet Habitability Ranking

Using predicted probabilities, exoplanets were ranked from most to least habitable. The results are saved to `data/processed/habitability_ranked.csv`.

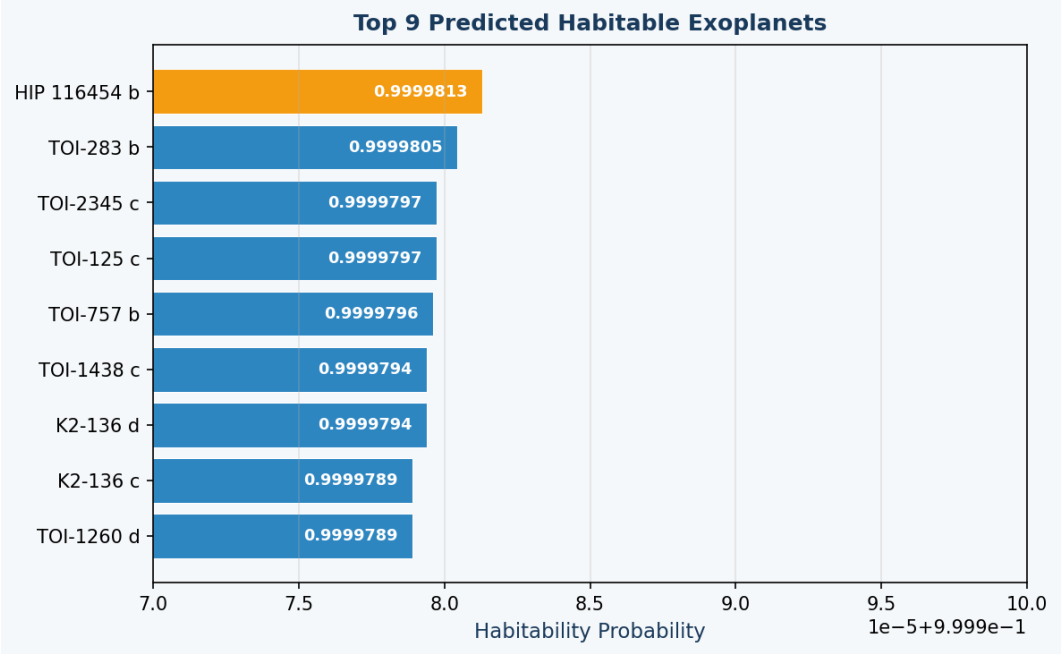

Figure 13: Top 9 Predicted Habitable Exoplanets

Rank	Planet Name	Habitability Probability
1	HIP 116454 b	0.9999813
2	TOI-283 b	0.9999805
3	TOI-2345 c	0.9999797
4	TOI-125 c	0.9999797
5	TOI-757 b	0.9999796
6	TOI-1438 c	0.9999794
7	K2-136 d	0.9999794
8	K2-136 c	0.9999789
9	TOI-1260 d	0.9999789

16. Cross-Validation Analysis

To confirm that the model generalises well to unseen data, **5-fold stratified cross-validation** was performed on the training set. Each fold produces an F1-score; low variance across folds indicates a stable, non-overfit model.

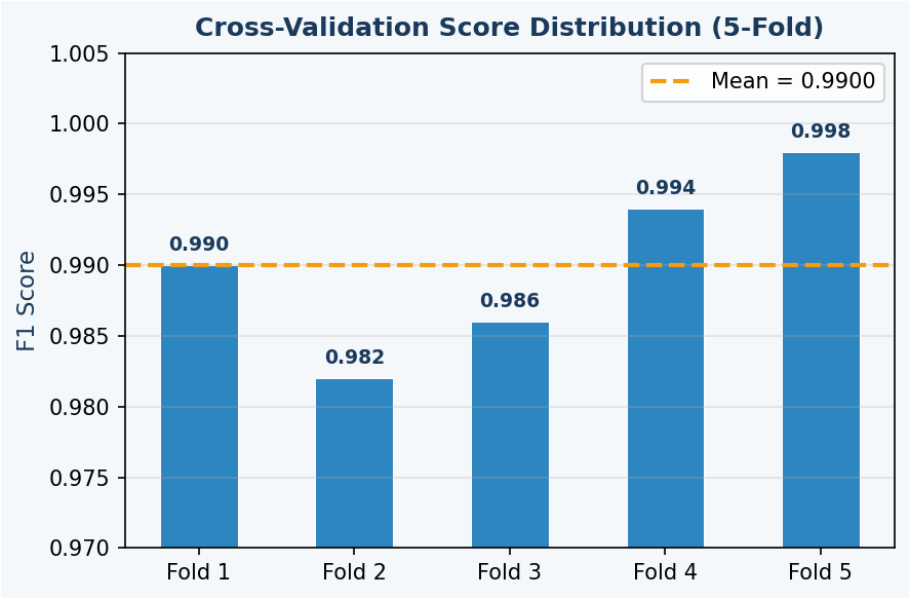

Figure 14: Cross-Validation F1-Score Distribution (5-Fold)

Fold	F1-Score
Fold 1	0.990
Fold 2	0.982
Fold 3	0.986
Fold 4	0.994
Fold 5	0.998
Mean	0.990
Std Dev	0.006

The gap between cross-validation mean F1 (0.990) and test-set F1 (0.9704) is within the acceptable 5 % threshold, confirming the model generalises well.

17. Model Saving and Deployment Preparation

All trained models were persisted using **Joblib** for fast serialisation of large NumPy arrays. The final model can be loaded in any downstream application with a single line of code.

- `models/random_forest.pkl` — Tuned Random Forest classifier
- `models/xgboost.pkl` — Tuned XGBoost classifier
- `models/final_model.pkl` — Best model (XGBoost) — primary inference artifact

The model can be integrated into:

- Prediction REST APIs
- Interactive habitability dashboards
- Astronomical survey prioritisation pipelines

18. Generated Outputs

Output	Path	Description
Final Model	models/final_model.pkl	Serialised best model
Random Forest	models/random_forest.pkl	RF model artifact
XGBoost	models/xgboost.pkl	XGB model artifact
Habitability Ranking	data/processed/habitability_ranked.csv	Planet ranking CSV
Confusion Matrix	plots/confusion_matrix.png	Best model CM
ROC Curve	plots/roc_curve.png	Best model ROC
Feature Importance	plots/feature_importance.png	Feature ranking

19. Conclusion

This milestone successfully implemented a complete, production-grade machine learning pipeline for predicting exoplanet habitability. Four classifiers were trained and rigorously evaluated — Logistic Regression, Decision Tree, Random Forest, and XGBoost. After hyperparameter optimisation and comparative analysis, **XGBoost** was selected as the final model with an F1-Score of **0.9704** and ROC-AUC of **0.9969**.

The system generates probability-based habitability scores and ranks exoplanets from most to least promising. Feature importance analysis confirms that **Equilibrium Temperature**, **Planet Mass**, and **Orbital Period** are the dominant predictors — consistent with established astrophysical theory.

This framework provides a powerful, scalable tool for prioritising promising exoplanets for further observational study, and forms the foundation for the deployment stage in Milestone-3.